ANSWER KEY — Integration Lesson 1 Test: Adding Up Little Pieces

For the grader · full working shown · give credit for correct method with small arithmetic slips

1. A train travels at a steady 80 miles per hour for 3 hours. How far does it go?

> **Work:** steady rate → distance = rate × time = 80 × 3. Units: (miles/hour) × hours = miles. **Answer: 240 miles.**

2. A faucet fills a tub at 6 liters per minute for 12 minutes. How much water is in the tub?

> **Work:** total = rate × time = 6 × 12 = 72. **Answer: 72 liters.**

3. A turtle walks 36 meters in 9 minutes at a perfectly steady pace. What is its speed in meters per minute?

> **Work:** this is the reverse question (rate from total): rate = total ÷ time = 36 ÷ 9 = 4. **Answer: 4 meters per minute.**

4. You bike at 10 mph for 2 hours, then at 6 mph for 3 hours. What is your total distance?

> **Work:** chop–multiply–add: 10 × 2 = 20 miles; 6 × 3 = 18 miles; total = 20 + 18. **Answer: 38 miles.**

5. A truck drives 40 mph for 1 hour, 60 mph for 2 hours, and 30 mph for half an hour. Find the total distance.

> **Work:** 40 × 1 = 40; 60 × 2 = 120; 30 × 0.5 = 15; total = 40 + 120 + 15. **Answer: 175 miles.**

6. A speed graph shows one flat step: 25 km/h from t = 0 to t = 4 hours. What is the area of the rectangle under the step, and what does that area mean in real life?

> **Work:** area = width × height = 4 × 25 = 100. Units: hours × km/hour = km. The area under a rate graph IS the accumulated total. **Answer: area = 100, meaning the object traveled 100 kilometers.**

7. A speed graph is a staircase: 6 m/s from t = 0 to t = 3 seconds, then 9 m/s from t = 3 to t = 8 seconds. Use rectangle areas to find the total distance.

> **Work:** rectangle 1: width 3, height 6 → 18 m. Rectangle 2: width 8 − 3 = 5 (NOT 8), height 9 → 45 m. Total = 18 + 45. **Answer: 63 meters.** (Common error: using width 8 for the second rectangle → 72 + 18 = 90, wrong.)

8. A jet flies at 600 miles per hour for 45 minutes. How far does it fly?

> **Work:** convert first: 45 minutes = 45/60 = 0.75 hours. Distance = 600 × 0.75 = 450. **Answer: 450 miles.** (600 × 45 = 27000 is the units-mixing trap.)

9. Water pours INTO a tank at 12 liters per minute while a drain removes 5 liters per minute. Both run for 8 minutes. How much water does the tank gain?

> **Work:** net rate = 12 − 5 = 7 L/min. Gain = 7 × 8 = 56. (Long way: 96 in − 40 out = 56 ✓.) **Answer: 56 liters.**

10. A runner wants to cover exactly 26 miles. She runs 8 mph for 2 hours, then 5 mph for 1 hour. She finishes the race at 5 mph. How many more hours must she run?

> **Work:** done so far: 8 × 2 = 16 and 5 × 1 = 5 → 21 miles. Remaining: 26 − 21 = 5 miles. At 5 mph: time = 5 ÷ 5 = 1. **Answer: 1 hour.**